

EXPERIMENT-34

Algorithm:

1. Start.
2. Display the menu:
Verify left distributive property.
Verify right distributive property.
Exit.
3. Read the user's choice and use a switch statement.
4. For left distributive property:
Read m, n, and p.
Input $A(m \times n)$, $B(n \times p)$ and $C(n \times p)$.
Calculate $B + C$.
Calculate the LHS: $A \times (B + C)$.
Calculate $A \times B$ and $A \times C$.
Calculate the RHS: $(A \times B) + (A \times C)$.
Display and compare LHS and RHS.
5. For right distributive property:
Read m, n, and p.
Input $A(m \times n)$, $B(m \times n)$ and $C(n \times p)$.
Calculate $A + B$.
Calculate the LHS: $(A + B) \times C$.
Calculate $A \times C$ and $B \times C$.
Calculate the RHS: $(A \times C) + (B \times C)$.
Display and compare LHS and RHS.
6. If corresponding elements of the LHS and RHS are equal, state that the selected property is verified.
7. Continue until the user selects Exit.
8. Stop.

Dimensions to remember:

Property	LHS	RHS	Final size
Left distributive	$A(B+C)$	$AB + AC$	$m \times p$
Right distributive	$(A+B)C$	$AC + BC$	$m \times p$

For both properties, the two final matrices should be identical.

Left: $A(B + C) = AB + AC$

Right: $(A + B)C = AC + BC$

Program, Compilation and Execution: